

Predicting the Emotional Impact of Videoclips

Luka Petrovic

Supervisor: Philipp Paukner

Chair for Data Processing, Technical University of Munich

luka.petrovic@tum.de

Abstract—Automatic video emotion recognition has gained increasing attention in affective computing, with the LIRIS-ACCEDE dataset being widely used as a benchmark [1]. While recent work has focused mainly on deep learning architectures such as CNNs, RNNs, or Transformers [2], feature-based models remain attractive due to their efficiency and interpretability. In this work, I investigate the use of LightGBM, a gradient boosted tree framework, for predicting valence and arousal from audiovisual descriptors.

Although recent neural network approaches report the highest scores, my LightGBM models achieve moderate but stable correlations on LIRIS-ACCEDE (up to $r \approx 0.28$ for valence and $r \approx 0.30$ for arousal), reflecting the challenging nature of continuous affect regression. A feature-importance analysis further reveals consistent modality-specific patterns: brightness- and color-based descriptors contribute most to valence, while spectral-envelope and motion-sensitive visual descriptors emerge as dominant predictors for arousal.

These findings highlight the value of interpretable, tree-based approaches as a complementary and transparent reference point for future research in video-level emotion recognition.

I. INTRODUCTION

A. Motivational Background

Automatic estimation of *viewer-induced* emotion from audiovisual media is a central topic in affective content analysis. Unlike traditional emotion recognition, which targets the emotional state of a person appearing in an image, video, or speech signal, viewer-induced emotion focuses on how audiovisual content elicits affective responses in the audience. This distinction is essential for applications in multimedia retrieval, film and advertisement analysis, and computational media design.

Whereas categorical emotion models assign discrete labels (e.g., happiness, anger), this affective approach uses a continuous dimensional model of affect, defined by *valence* (negative–positive) and *arousal* (inactive–active), rather than discrete emotion categories. One of the common benchmarks for this dimensional model is the LIRIS-ACCEDE dataset [1], which has emerged as a widely adopted benchmark for the prediction of *viewer-induced* emotion from audiovisual content. It comprises approximately 9,800 short video clips of 8–12s duration, sampled from 160 Creative Commons movies spanning a broad range of genres. Clips are extracted along shot or fade boundaries and preserve within-movie temporal order, yielding homogeneous excerpts suitable for affect elicitation.

Recent progress in viewer-induced emotion prediction has been driven largely by deep neural architectures such as CNNs,

RNNs, and Transformers [2], [3]. These systems can extract rich semantic representations but are computationally intensive and often lack interpretability, making it difficult to understand *why* a model predicts a particular affective response.

In contrast, classical feature-based approaches offer several advantages that remain relevant: (1) they provide transparent, interpretable mappings between audiovisual descriptors and affective outcomes, (2) they are lightweight and efficient, allowing rapid experimentation and deployment on limited hardware, and (3) they supply actionable insights for content creators, editors, and generative-media systems. For example, understanding how brightness, color harmony, motion intensity, or spectral energy contribute to elicited emotions can support emotion-driven video editing, targeted content design, or controllable generative models that seek to modulate viewer affect in a predictable way.

Tree ensembles such as gradient-boosted decision trees (GBDTs) enable direct inspection of feature importances and thus offer a principled way to analyze which cues most strongly influence predicted affective responses. In this study, interpretability and diagnostic value were prioritised over raw predictive accuracy, making GBDTs a suitable baseline for investigating viewer-induced emotion in an explainable and reproducible manner.

B. State of the Art in Video Emotion Recognition

The LIRIS-ACCEDE dataset was introduced by Baveye et al. [1] as a large-scale benchmark for induced emotion analysis in videos, with continuous valence and arousal as primary prediction targets. Early approaches on this dataset relied on handcrafted audiovisual descriptors combined with shallow learning models such as support-vector machines and late-fusion schemes.

Subsequent work increasingly adopted deep learning models to learn hierarchical representations from audiovisual data. CNN-based approaches using visual backbones such as VGG or ResNet, together with deep audio embeddings, consistently outperform handcrafted features when evaluated with regression metrics [3], [4], [5]. Across MediaEval-based evaluations on LIRIS-ACCEDE, such convolutional or hybrid deep models typically report MSE values in the range of approximately 0.09–0.11 for valence and 0.13–0.15 for arousal, indicating a substantial improvement over shallow baselines but still leaving considerable room for further gains, especially for valence.

More recent state-of-the-art methods focus on explicitly modeling temporal structure and long-range dependencies in

video. Multimodal fusion architectures with shot-level context, recurrent components, or attention mechanisms achieve consistently lower errors on LIRIS-ACCEDE [2], [6]. A comprehensive comparison of regression-based approaches is provided by Mittal et al. [7], who report MSE values spanning approximately 0.07–0.10 for valence and 0.13–0.14 for arousal across a wide range of competitive models evaluated under a unified protocol. Among these, multimodal models with explicit temporal context achieve the lowest reported errors, with valence remaining consistently more challenging than arousal.

At the upper end of current performance, Zhao et al. [6] report valence MSE values close to 0.07 and arousal MSE around 0.14 using multimodal fusion with shot-level temporal modeling, representing a strong reference point for regression-based affect prediction on LIRIS-ACCEDE. Similar trends are observed in recent correlation-aware and attention-based architectures [2], [3], which further reduce error by capturing long-range temporal and cross-modal dependencies.

Despite these advances, recent surveys emphasize that increasing predictive performance is typically accompanied by substantially higher model complexity and reduced interpretability [3], [5], [8].

C. Problem Statement and Research Objectives

Despite the recent advances in deep learning for video-based affect analysis, transparent and interpretable baselines for continuous valence–arousal prediction under strict generalization constraints remain underexplored. In particular, it is still unclear how well feature-based, tree-ensemble models generalize across unseen movies and which audio–visual cues they rely on when predicting induced emotion.

This work addresses this gap by establishing an interpretable, tree-based regression baseline using LightGBM [9]. The goal is not to close the performance gap to deep architectures, but to reduce the knowledge gap by providing a transparent reference point that allows detailed analysis of predictive behaviour, generalization, and feature contributions.

Scope. The scope of this study is limited to induced emotion prediction on the LIRIS-ACCEDE dataset using continuous valence and arousal annotations. All experiments follow a strict movie-disjoint protocol. The approach focuses on handcrafted audio–visual features and gradient-boosted decision trees, explicitly excluding categorical emotion labels, end-to-end deep learning architectures, and explicit temporal sequence models.

Research Questions.

- **RQ1 (Predictive performance):** How well do LightGBM regressors predict continuous valence and arousal on LIRIS-ACCEDE under a movie-disjoint protocol?
- **RQ2 (Generalization):** How robustly do feature-based LightGBM models generalize to unseen movies, and how does this behaviour differ between valence and arousal?
- **RQ3 (Feature contributions):** Which audio–visual descriptors contribute most to model predictions, and what do they reveal about modality salience in induced affect?

II. METHODOLOGY

A. Preprocessing and Feature Representation

All clips were assigned to three strictly *movie-disjoint* splits (train, validation, test), ensuring that no film appears across multiple partitions. Feature extraction was performed using an extended version of the in-house *vFeat* framework, including reimplemented audio descriptors originally based on deprecated Yaafé components [10].

Each clip is represented by frame-level audiovisual descriptors, containing up to 768 temporal samples and 248 feature dimensions per frame, organized into semantically coherent groups (e.g., spectral, chroma, motion, and color descriptors). Group-wise normalization schemes were applied according to feature semantics, including **Min–Max**, **Log–Min–Max**, **Z-score**, and **STD**-based scaling. For composite groups (e.g., `magnitude_bins` or `mel_bands`), normalization statistics were computed jointly across all group members.

Since LightGBM requires fixed-length inputs, temporal feature sequences were summarized using statistical aggregates computed along the time axis (mean, standard deviation, extrema, median, energy, slope, skewness, kurtosis). Each clip is thus represented by a flattened vector of selected feature–aggregate pairs, normalized per feature group and free of missing values.

B. Gradient-Boosted Decision Trees and LightGBM Regression Framework

Gradient-Boosted Decision Trees (GBDTs) combine decision trees with additive ensemble learning by iteratively fitting weak learners to the residuals of previous predictions [11]. This makes GBDTs particularly suitable for tabular data such as aggregated audio–visual descriptors.

A regression tree partitions the input space into disjoint regions R_j and assigns a constant prediction b_j to each region [12]:

$$h(x) = \sum_{j=1}^J b_j \mathbf{1}\{x \in R_j\}. \quad (1)$$

GBDTs minimize a differentiable loss $L(y, F(x))$ by successively fitting trees to pseudo-residuals, defined as the negative gradient of the loss with respect to current predictions:

$$r_{im} = -\frac{\partial L(y_i, F_{m-1}(x_i))}{\partial F_{m-1}(x_i)}. \quad (2)$$

The model update follows

$$F_m(x) = F_{m-1}(x) + \nu \rho_m h_m(x), \quad (3)$$

where ν denotes the learning rate and ρ_m the optimal step size.

The regression framework was implemented using LightGBM [9] and a custom wrapper that trains two independent regressors—one for valence and one for arousal—under identical hyperparameter settings. Hyperparameters include learning rate, number of estimators, tree depth, leaf count, and L_1/L_2 regularization.

Interpretability is preserved through built-in LightGBM importance measures. *Split importance* reflects feature usage frequency, whereas *gain importance* captures total loss reduction attributed to each feature [9]. These measures form the basis for subsequent analysis of modality-specific predictors.

LightGBM extends the standard GBDT framework with several optimizations that make it scalable and robust [9]. Key mechanisms include histogram-based binning for faster split search, gradient-based one-side sampling (GOSS) for focusing on informative samples, and leaf-wise growth that adaptively deepens the most promising branches.

C. Feature Selection Strategy

To reduce redundancy and improve interpretability, a structured feature selection pipeline was applied prior to model training, following established filter-based selection principles [13]. All steps were implemented in a custom Python framework built around LightGBM regressors and cross-validation.

a) Selection pipeline.: Starting from the aggregated representations described in Section II-A, features were processed in two disjoint subsets: *composite* descriptors (e.g., spectral or chroma features) and *non-composite* atomic features (e.g., brightness or motion continuity). The selection consisted of five consecutive stages:

- 1) **Variance filtering**: Near-constant features ($< 10^{-10}$) were removed.
- 2) **Mutual information ranking**: The top- k features ($k = 300$) were retained per target.
- 3) **Cross-validated importance estimation**: Gain- and split-based LightGBM importances were aggregated across 10-fold cross-validation and multiple random seeds.
- 4) **Permutation validation**: Permutation scores were used to verify robustness of relevance rankings.
- 5) **Redundancy pruning**: Highly correlated features (Spearman $|\rho| > 0.9$) were iteratively removed.

b) Category-aware export.: Selected features were exported into structured JSON configurations for downstream training, separately for valence and arousal and in multiple subset sizes. In addition to global selections, category-specific variants (e.g., *color_light*, *spectral_shape*, *temporal_rhythm*) were generated to enable focused ablation studies. Each configuration fully specifies the feature bases and aggregates used in subsequent experiments.

III. EXPERIMENTS

A. Model Setup and Training

The selected feature subsets were used to define a series of LightGBM regression experiments. Each experiment corresponds to a *schedule*, specifying the combination of feature sets, aggregate statistics, and hyperparameters to be trained. Schedules were generated automatically via a Python utility, which assembled reproducible configuration lists stored in JSON format.

a) Schedule generation.: Each schedule defines a compact set of hyperparameters and feature combinations explored systematically across targets. Hyperparameters include learning rate (0.02–0.1), number of estimators (800–2400), maximum tree depth (6–10 or unrestricted), and leaf count (31–127). In addition, schedules specify the aggregate statistics (e.g., mean, standard deviation, extrema, energy, skewness, kurtosis) and the selected audio–visual feature groups (e.g., *temporal rhythm*, *color light*, *spectral shape*, *motion dynamics*). This design enables consistent coverage of diverse low-, mid-, and high-level descriptor combinations across both modalities.

b) Training and evaluation.: All schedules were trained independently using a fixed random seed to ensure reproducibility and evaluated on the movie-disjoint validation split. Performance metrics (MSE, RMSE, Pearson r , and R^2) were computed separately for valence and arousal. After training, both split- and gain-based feature importances were extracted from each regressor to quantify the contribution of individual features and feature groups.

B. Results and Generalization Analysis

To assess the predictive behaviour of the LightGBM regressors, all scheduled models were evaluated under the strict *movie-disjoint* protocol for both *valence* and *arousal*. During the large-scale screening phase, performance was measured on the normalized target range $[-1, 1]$, which stabilizes optimization across configurations. Pearson correlation is invariant under this scaling, whereas MSE, RMSE, and R^2 reflect the respective label domain.

a) Validation performance.: Across the validation split, performance distributions are stable and moderately predictive. Evaluation metrics show generally stronger results for large-feature configurations. The highest-scoring models from the screening phase are summarized in Table I, reported on the normalized scale and used solely for comparative ranking within the screening stage.

TABLE I. BEST VALIDATION MODELS PER TARGET AND CAPACITY GROUP (NORMALIZED LABEL SCALE).

Model (ID)	Target	$ F $	RMSE	MSE	r	R^2
Arousal Large (lgbmuy-419a652)	Arousal	144	0.4609	0.2126	0.3086	0.0887
Arousal Small (lgbmeh-eea06c1)	Arousal	48	0.4716	0.2224	0.2608	0.0460
Valence Large (lgbmv6-84f7612)	Valence	143	0.3090	0.0955	0.2843	0.0693
Valence Small (lgbmfp-78d3c21)	Valence	48	0.3137	0.0984	0.2589	0.0407

b) Evaluation on the original label scale.: To provide interpretable error magnitudes, the four top-performing models (one per target and feature-size group) were re-evaluated on the dataset’s native label range $[1, 5]$. Table II reports validation and test performance for these models. Validation correlations match the screening trends. Under the movie-disjoint evaluation, test-set performance reveals target-dependent generalization behavior: arousal models retain stable or slightly improved

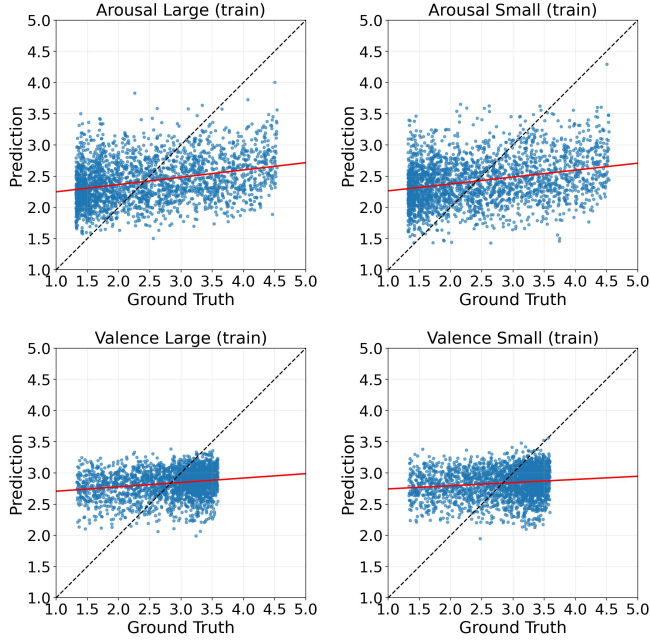


Fig. 1. Scatterplots for the four top models trained on the *training split* and evaluated on the *test split*. Panels are ordered row-wise as follows: **Arousal Large** (top left), **Arousal Small** (top right), **Valence Large** (bottom left), and **Valence Small** (bottom right).

correlations across splits, whereas valence models exhibit a clearer reduction from validation to test.

A visual inspection of the prediction behaviour is provided in Fig. 1. All models show weak but consistent positive linear trends, with slightly steeper slopes for the large-feature configurations. Predictions remain concentrated around a narrow band, reflecting the limited dynamic range and skewness of the target distributions.

TABLE II. FINAL EVALUATION OF THE TOP FOUR MODELS ON THE ORIGINAL LABEL SCALE (TRAIN $\rightarrow$ VAL AND TRAIN $\rightarrow$ TEST).

Model (ID)	Split	Target	RMSE	MSE	r	R^2
Arousal Large (lgbmuy-419a652)	Val	Arousal	0.9219	0.8498	0.3086	0.0887
		Arousal	0.9067	0.8221	0.3106	0.0912
Arousal Small (lgbmeh-eea06c1)	Val	Arousal	0.9432	0.8897	0.2608	0.0460
		Arousal	0.9192	0.8449	0.2815	0.0661
Valence Large (lgbmv6-84f7612)	Val	Valence	0.6180	0.3819	0.2843	0.0693
		Valence	0.6108	0.3730	0.1976	0.0140
Valence Small (lgbmfp-78d3c21)	Val	Valence	0.6275	0.3937	0.2589	0.0407
		Valence	0.6294	0.3962	0.1313	-0.0472

c) *Extended training and generalization*: To assess whether additional training data improve generalization, the same four models were retrained on the combined Train+Validation split and evaluated on the held-out test set. Results are summarized in Table III. Extended training yields target-dependent effects, with clear improvements for arousal, while valence gains remain limited or inconsistent. The corresponding scatterplots in Fig. 2 show slightly tighter point

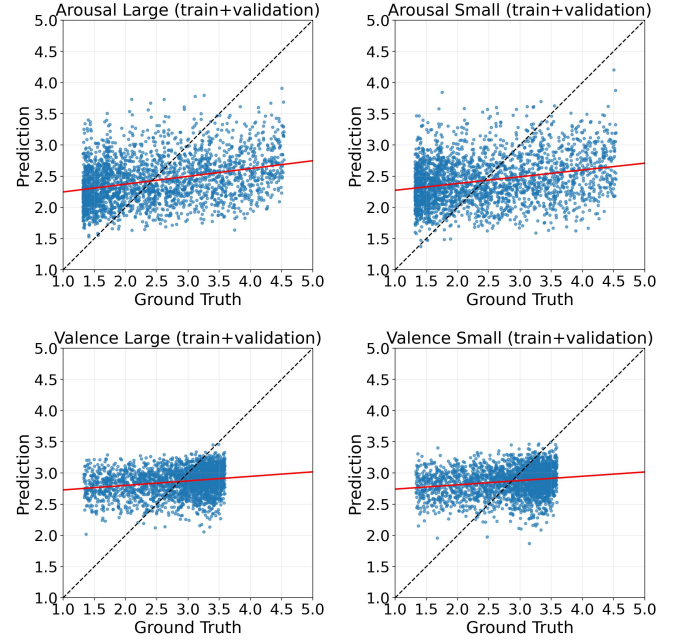


Fig. 2. Scatterplots for the four top models trained on the *combined Train+Validation split* and evaluated on the *test split*. Panels are ordered row-wise as follows: **Arousal Large** (top left), **Arousal Small** (top right), **Valence Large** (bottom left), and **Valence Small** (bottom right). The regression lines illustrate the linear trend for each model, with slightly stronger alignment visible for the arousal regressors.

clouds for arousal models, whereas changes for valence are less pronounced, consistent with the numerical trends.

TABLE III. EXTENDED TRAINING RESULTS (TRAIN+VAL $\rightarrow$ TEST) ON THE ORIGINAL LABEL SCALE.

Model (ID)	Target	RMSE	MSE	r	R^2
Arousal Large (lgbmuy-419a652-final)	Arousal	0.9046	0.8183	0.3185	0.0955
Arousal Small (lgbmeh-eea06c1-final)	Arousal	0.9248	0.8553	0.2695	0.0546
Valence Large (lgbmv6-84f7612-final)	Valence	0.6078	0.3694	0.2098	0.0235
Valence Small (lgbmfp-78d3c21-final)	Valence	0.6153	0.3785	0.1872	-0.0005

d) *Overfitting behaviour*: Comparing validation and test performance reveals a clear difference between target dimensions. Valence models exhibit moderate overfitting, with noticeable performance drops when evaluated on unseen movies, suggesting that valence-related visual cues are less consistent across films. In contrast, arousal generalizes substantially better: correlations remain stable or even improve slightly across splits, including after extended training. This pattern indicates that the descriptors driving arousal—such as motion-sensitive and activity-related features—transfer more reliably across movies (see Section III-C).

Overall, the results suggest that the LightGBM regressors are limited more by representational capacity than by excessive variance. While valence prediction remains challenging under

the movie-disjoint protocol, arousal shows comparatively robust generalization, and extended training yields controlled, target-specific improvements without indications of severe overfitting.

C. Feature Importances and Interpretability

To identify which descriptors contributed most consistently to affective prediction, feature importances were analyzed across the top 10% of all screening-phase models per target dimension. Model selection was based on a two-dimensional filtering that retained configurations closest to the ideal point defined by minimal normalized MSE and maximal Pearson r , thereby capturing models that jointly exhibit low error and strong linear association. These selected models represent the most stable and well-performing configurations within the screening phase.

a) Dominant features for valence: Valence predictions are dominated by visual descriptors related to color, brightness, and overall image composition. Across the selected high-performing models, aggregated measures of color energy, color activity, and brightness consistently rank among the most influential features (Fig. 3). These findings indicate that static visual characteristics exert a stronger influence on perceived valence than temporal structure. Audio-derived descriptors contribute only secondarily, and motion-related features show consistently low importance, suggesting limited relevance of temporal dynamics for valence prediction under the movie-disjoint protocol.

b) Dominant features for arousal: In contrast to valence, arousal importance patterns reflect a broader mixture of feature groups. Highly ranked descriptors include aggregates capturing spectral-envelope characteristics, spatial composition, and low-level activity cues (Fig. 4). Rather than being driven by a single energetic feature family, arousal predictions appear to integrate multiple indicators of variability and intensity, including motion-sensitive visual statistics and spectral shape proxies. This distribution is consistent with the notion that arousal reflects temporal and structural activation rather than purely static content.

c) Split-based patterns: Split-based importances are more equalized than their Gain-based counterparts, with mid-ranked features appearing frequently even when their absolute contributions are moderate. Nonetheless, several descriptor groups—most notably spatial composition measures, color activity statistics, and selected spectral descriptors—emerge consistently across both importance metrics. This overlap suggests that the identified patterns are robust to the specific importance aggregation scheme and reflect stable predictive relevance.

IV. DISCUSSION

RQ1 (Predictive performance). Across the validation split, the proposed models achieve peak correlations of approximately $r \approx 0.28$ for valence and $r \approx 0.30$ for arousal (cf. Section III-B), which is consistent with expectations for feature-based, interpretable regression approaches on LIRIS-ACCEDE. These results underline the challenging nature of

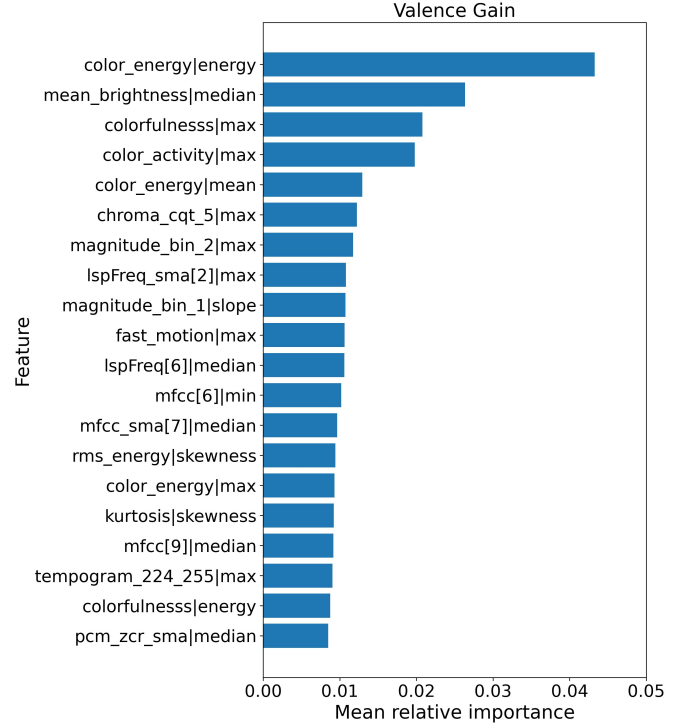


Fig. 3. Top features ranked by mean Gain-based importance across the selected Valence models.

continuous affect prediction on this dataset, particularly for valence. Recent deep-learning approaches reviewed by Guo et al. [5] and Yi et al. [2] typically report higher correlations in the range $r = 0.35$ – 0.45 , albeit at the cost of substantially increased model complexity and reduced interpretability. In terms of error-based metrics, however, the achieved MSE values for both valence and arousal remain noticeably above those reported by recent state-of-the-art approaches (cf. Section I-B), indicating a clear performance gap in absolute prediction accuracy.

RQ2 (Generalization). Evaluation on held-out movies reveals a clear difference in generalization behaviour between the two affective dimensions. While arousal models retain stable correlations when moving from validation to test data, valence models exhibit a more pronounced performance drop, indicating moderate overfitting under the movie-disjoint protocol. This suggests that arousal-related cues generalize more reliably across films, whereas valence prediction depends more strongly on global visual context that varies between movies.

RQ3 (Feature contributions). Beyond predictive accuracy, the analysis of feature importances reveals modality-specific patterns that are broadly consistent with prior findings in affective computing (Section III-C). For valence, predictions are dominated by static visual cues related to brightness, color, and global composition, in line with well-established brightness–valence relationships summarized by Guo et al. [5]. In contrast, arousal reflects a broader mixture of activity- and structure-related cues, consistent with the emphasis on temporal and

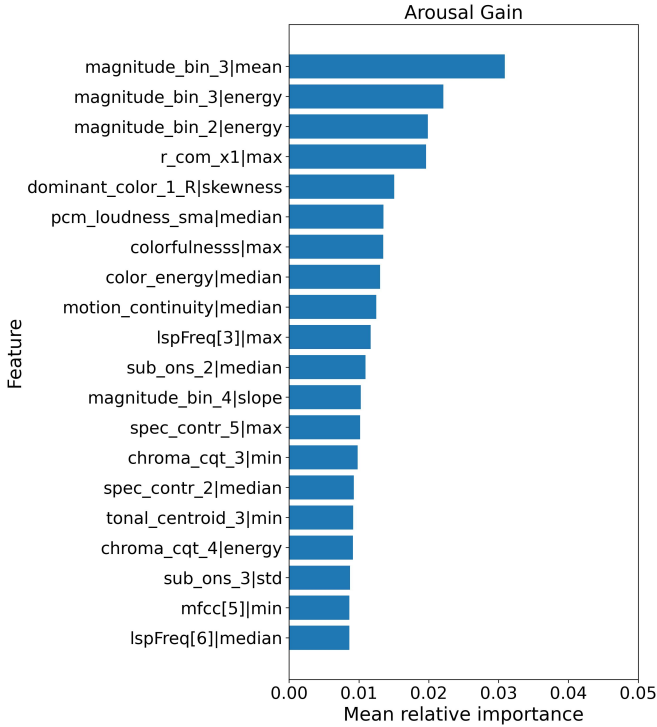


Fig. 4. Top features ranked by mean Gain-based importance across the selected Arousal models.

energetic dynamics reported by Yi et al. [2]. These results show that the LightGBM models recover modality-specific regularities that closely mirror those identified by more complex deep architectures, supporting their use as interpretable baselines for analyzing audiovisual correlates of induced affect.

V. CONCLUSION

This work presented an interpretable and computationally efficient baseline for continuous valence–arousal prediction on the LIRIS-ACCEDE dataset. The experimental results confirm well-known trends in video emotion recognition: arousal generalizes more robustly across unseen movies than valence, while valence prediction remains challenging due to its stronger dependence on global visual context. Although the proposed models do not reach the predictive accuracy of modern deep architectures, they provide a transparent and reproducible reference point that enables direct inspection of modality-specific feature contributions.

Several limitations remain. The current approach relies on static aggregation of handcrafted descriptors and therefore does not explicitly model temporal affect dynamics within clips. In addition, large-scale screening was restricted to the validation split due to computational constraints, and the feature space may overlook higher-level semantic cues captured by end-to-end deep representations.

Future work will therefore focus on extending the baseline in two directions. First, simple temporal or hierarchical aggregation mechanisms could be integrated to bridge the gap between

static feature pooling and sequential modeling while preserving interpretability. Second, combining interpretable GBDTs with pretrained audio–visual embeddings (e.g., from VGGish or CLIP) may enable hybrid models that retain diagnostic transparency while capturing higher-level semantic information.

Overall, this study establishes an interpretable and computationally lightweight baseline for continuous affect prediction on LIRIS-ACCEDE and highlights how characteristic audio–visual patterns underlying induced emotion can be analyzed in a transparent and reproducible manner.

REFERENCES

- [1] Y. Baveye, J.-N. Bettinelli, E. Dellandréa, L. Chen, and C. Chamaret, “A large video database for computational models of induced emotion,” in *2013 Humaine Association Conference on Affective Computing and Intelligent Interaction*, 2013, pp. 13–18. doi: 10.1109/ACII.2013.9.
- [2] Y. Yi, J. Zhou, H. Wang, P. Tang, and M. Wang, “Emotion recognition in user-generated videos with long-range correlation-aware network,” *IET Image Processing*, vol. 18, pp. 3288–3301, 07 2024. doi: 10.1049/ipr2.13174.
- [3] Y. Yi, Y. Zhou, T. Wang, and J. Zhou, “Advances in video emotion recognition: Challenges and trends,” *Sensors*, vol. 25, no. 12, 2025. doi: 10.3390/s25123615.
- [4] Y. Yi, H. Wang, and Q. Li, “Cnn features for emotional impact of movies task,” in *Proceedings of the MediaEval 2019 Workshop*, Sophia Antipolis, France, Sep. 2019. [Online]. Available: https://ceur-ws.org/Vol-2283/MediaEval_18_paper_13.pdf
- [5] J. Xue, J. Wang, X. Liu, Q. Zhang, and X. Wu, “Affective video content analysis: Decade review and new perspectives,” *Big Data Mining and Analytics*, vol. 8, no. 1, pp. 118–144, 2025. doi: 10.26599/BDMA.2024.9020048.
- [6] J. Zhang, Y. Zhao, L. Cai, C. Tu, and W. Wei, “Video affective effects prediction with multi-modal fusion and shot-long temporal context,” *arXiv preprint arXiv:1909.01763*, Sep. 2019. doi: 10.48550/arXiv.1909.01763.
- [7] T. Mittal, P. Mathur, A. Bera, and D. Manocha, “Affect2mm: Affective analysis of multimedia content using emotion causality,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 5657–5667. doi: 10.1109/CVPR46437.2021.00561.
- [8] A. Antonov, S. Sampath Kumar, J. Wei, W. C. Headley, O. Wood, and G. Montana, “Decoding viewer emotions in video ads,” *Scientific Reports*, vol. 14, p. 26382, 11 2024. doi: 10.1038/s41598-024-76968-9.
- [9] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “Lightgbm: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. doi: 10.48550/arXiv.1711.08788.
- [10] B. Mathieu, S. Essid, T. Fillon, J. Prado, and G. Richard, “Yaaf, an easy to use and efficient audio feature extraction software,” in *Proceedings of the 11th International Society for Music Information Retrieval Conference (ISMIR)*, Utrecht, Netherlands, 2010, pp. 441–446. [Online]. Available: <https://ismir2010.ismir.net/proceedings/ismir2010-75.pdf>
- [11] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001. doi: 10.1214/aos/1013203451.
- [12] L. Breiman, J. Friedman, R. A. Olshen, and C. J. Stone, *Classification and Regression Trees*. Chapman & Hall/CRC, 1984. doi: 10.1201/9781315139470.
- [13] I. Guyon and A. Elisseeff, “An introduction of variable and feature selection,” *J. Machine Learning Research Special Issue on Variable and Feature Selection*, vol. 3, pp. 1157–1182, 01 2003. doi: 10.1162/153244303322753616.